1. **One-dimensional statistical diagrams** such as line graph, poly graph, bar diagram, histogram, and star diagram. etc., in these diagrams quantities are represented by the line, length of bars and length of radiating lines in a star diagram.
2. **Two-dimensional statistical diagrams** are also known as area diagram such as squares, rectangles and circles. Divided squares, divided circles also called as pie diagrams are two dimensional diagrams.
3. **Three-dimensional statistical diagrams** are also called volume diagram examples cube, block piles and spherical diagrams.

We will discuss the following diagrams alone

Line graphs

Bar diagrams

Pie diagram

12.2.1 Line Graph

The line graphs are usually drawn to represent the time series data related to the temperature, rainfall, population, birth rates and the death rates.

Uses of line-graphs

Line-graphs are commonly used for representing

- Climatic data pertaining to elements such as temperature, pressure, humidity, etc.,
- Population data such as growth of population, birth rate, death rate, etc, and
- Economic data such as crop yield, minerals extraction, production of industrial goods over a period of time etc.

Simple line graph

A simple line-graph is a line or a curve drawn by joining together, points plotted with the help of two co-ordinates i.e. is a vertical axis and horizontal scale. It represents the successive changes in the value of one variable over a certain period of time.

How to construct a Line Graph

Line graphs are very easy to construct on graph paper.

1. Draw a horizontal line to represent x-axis and vertical line to represent y-axis. The point where x-axis and y-axis intersect at right angles is the starting point. Normally assigned the value zero (0).
2. Next add label and scale for both axes.
3. After that, plot your data points.
4. Next draw a line through the plotted data points.
5. Finally add a title.

Let us see some of the solved examples given below.

Example 12.1

Construct a line graph to represent the data as given in Table 12.1

Table 12.1: Literacy Rate of India – 1901 to 2011

Year	1901	1911	1921	1931	1941	1951	1961	1971	1981	1991	2001	2011
Literacy rate in %	5.35	5.92	7.16	9.50	16.10	16.67	24.0	29.45	36.23	42.23	64.83	74.04

Source: Census report 2011

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